WASP-3b

R-1062

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-3_150528` · created 2026-07-30T12:00:49+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457170.8171 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.0905 +/- 0.0078
Transit depth (Rp/Rs)^2	0.82 +/- 0.14 [%]
Orbital Inclination (inc)	84.1 +/- 1.24
Airmass coefficient 1 (a1)	1.0527 +/- 0.0077
Airmass coefficient 2 (a2)	-0.0429 +/- 0.004
Scatter in the residuals of the lightcurve fit is	0.73 %
Best Comparison Star	#2 - [290, 176]
Optimal Aperture	3.6
Optimal Annulus	11.67
Transit Duration (day)	0.1132 +/- 0.0067

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.0905 ± 0.0078 vs 0.10538 (deviation 1.08σ)
Duration fit vs published	0.1132 d vs 0.1167 d (deviation 0.29σ)
Mid-time O–C	4.5 min
Depth SNR	6.6
Residual scatter	0.73 %

Light curve

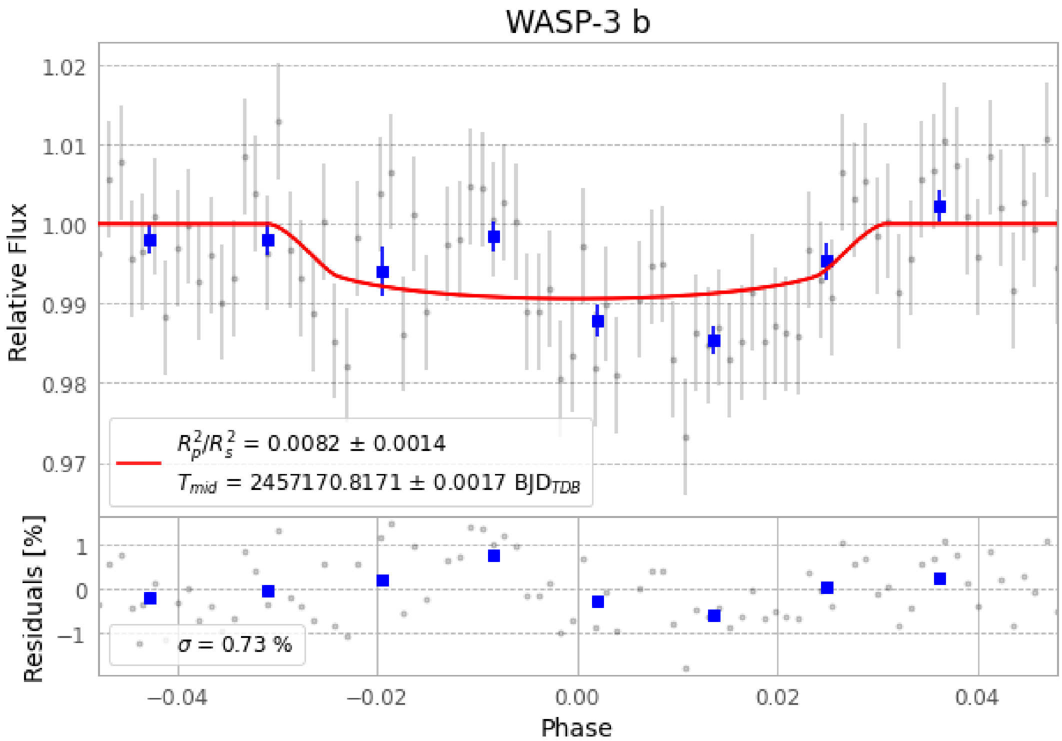

Figure 1 — FinalLightCurve_WASP-3 b_2015-05-27.png (SHA-256 567a3a90e6201a73…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [217, 226], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 0ae2cbf96d2921b8…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

86 FITS frames (56 MB), `WASP-3150528052412.FITS` → `WASP-3150528093912.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-150528.log (163ec14c0054…), FinalLightCurve_WASP-3 b_2015-05-27.png (567a3a90e620…), FinalLightCurve_WASP-3 b_2015-05-27.csv (542efc921021…)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)